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CPT_S 440 Spring 2026

Pokemon Red Reinforcement Learning Report

Project Goals:

- To install and run the Pokemon Red training program from Whidden's GitHub.
- To build upon the existing reinforcement training and improve the success rate of completing Mt. Moon in the original program format.
- Originally, more heuristic data was going to be used. I decided to do more reshaping of the rewards and penalties than heuristics as the project progressed.

Project Methods:

- Heuristic data on speed run times was collected from a YouTube speedrun video, @pokeguy's video: "Pokemon Red Any% Glitchless Speedrun in 1:44:03 [Former World Record]" at <https://www.youtube.com/watch?v=MSOZzdIIN4A>. This data was used to create a tiered reward system for getting to certain maps, like Route 1 and Viridian City in close to speedrun times, and a lower reward for double the speedrun time. This is still part of the reward system, so it is really still re-shaping.
- Penalties were added for remaining in the menu and in one place for too long, and for remaining on the same map for too long. These are hard values set high to not punish NPC dialogue.
- A movement penalty was added for jittering, so that the AI would not switch back and forth between buttons to simulate movement but go nowhere.
- The reward system for distance is unfortunately in need of repair as of writing this. Instead of rewarding steps in the same direction, it is rewarding not changing direction, even if without movement. The AI will farm anything low -risk and low – effort, and it has been farming this. The rewards have been nerfed, but if needs to be re-worked to only reward the same direction is there is also a change in the (x,y) position. Because one of the biggest Mt. Moon obstacles is walking long distances with no reward, it needs to be reworked and not removed.
- The item reward is only for free items. If it applied to the shop, the AI would immediately spend all its money on something random. This is also because the AI must pick up the Fossil item to be able to leave Mt. Moon.
- The HM and badge rewards are to encourage game progression. Sadly, the AI never made it that far, so they are yet untested.

- The combat rewards are for high-damage hits and enemy pokemon KOs. In mid-development, these rewards were set too high. The AI farmed the grass patch above the starting town for points and made no further progress. They have since been nerfed.
- The original healing reward could be farmed for points by getting injured and then going to heal in a loop. The reward was nerfed, but this never seemed to be an issue. It was a possibility that the AI never tried.
- The exploration reward was nerfed from the original value for fear that it would interfere with the other rewards. Progression through the game came to a grinding halt. If anything, it needs to be increased to stay the driving force and encourage moving to new areas to discover the other rewards. It has been increased, and probably needs to increase until it is the single greatest reward. It rewards for each step on new (x,y) territory.
- A reward was added for adding pokemon to the party. This is not strictly necessary for game progression, but the AI wanders around and battles and faints often in Mt. Moon. Therefore, a larger party would aid in game progression through the cave in particular.

Project Results:

Sadly, I did not get the completion of Mt. Moon that I wanted. The rewards and penalties that I added often did not work, broke other existing code, or over-shadowed exploration which is essential to completing the game. I believe the only changes needed now are to increase exploration slightly and fix the direction reward to only work for moving players. There may be other problems in the reward system, but these are the problems preventing progression right now. I will likely alter them, and train the AI again after the semester is done.

This kind of PPO reward training is time-consuming, especially on the old, more inefficient settings that work on my temperamental old computer. That said, I am not using it for anything else, so I want to continue to experiment with the reward shaping.